

Architecture: Core Banking Integration Layer

Meridian Commercial Bank - Internal Document

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Overview

The Core Banking Integration Layer (CBIL) sits between customer-facing channels (web, mobile, branch systems) and the core ledger system of record. It exists to decouple channel teams from the core ledger's legacy batch-oriented interface, exposing a modern async API surface instead.

Components

- Ledger Adapter: Translates async API calls into the core ledger's native batch file format, queues them, and polls for completion.
- Balance Cache: A near-real-time read replica of account balances, refreshed via change-data-capture from the core ledger, used to serve low-latency balance inquiries without hitting the ledger directly.
- Event Publisher: Publishes account and transaction events (deposit, withdrawal, hold placed/released) to an internal event bus consumed by downstream systems such as fraud scoring and customer notifications.

Design Rationale

The core ledger system processes updates in batch windows every 15 minutes, which is incompatible with customer expectations of near-instant balance updates. The Balance Cache exists specifically to bridge this gap: writes go through the Ledger Adapter (eventually consistent, batch-backed), while reads are served from the cache (near-real-time, seconds of lag via CDC) for a responsive user experience.

Consistency Model

CBIL is eventually consistent by design. The Balance Cache may lag the authoritative ledger by up to 15 seconds under normal operation. Any feature requiring strong consistency (e.g., final balance check immediately before a large wire transfer) must read directly from the Ledger Adapter's synchronous confirmation path, not the Balance Cache.

Failure Modes

- If CDC lag exceeds 60 seconds, the Balance Cache is marked stale and read requests are transparently routed to the synchronous Ledger Adapter path (higher latency but always fresh).
- If the Ledger Adapter's batch queue backs up, writes are still accepted and queued; customers see a "pending" transaction state rather than a failure.

Related Documents

- ARCH-payments-platform-overview (Payments Platform is a consumer of CBIL's Ledger Adapter for settlement)

- ARCH-data-platform (Event Publisher feeds the Data Platform's event ingestion pipeline)